Table 5: Retrieval Ablation Results across seven configurations (20 ground-truth queries). † All configurations use character 3-gram cosine similarity as the dense retrieval proxy; the deployed production system uses Pinecone sentence embeddings. These results therefore characterise the offline surrogate pipeline.

Configuration	**R@5**	**R@10**	**P@5**	**MRR**	**nDCG@5**	**nDCG@10**
Flat Lexical Only	45.0%	50.0%	9.0%	0.384	0.394	0.411
Flat Dense Only	35.0%	45.0%	7.0%	0.281	0.288	0.322
Flat Hybrid (Score Sum)	45.0%	50.0%	9.0%	0.384	0.397	0.411
Flat Hybrid (RRF)	40.0%	50.0%	8.0%	0.413	0.400	0.432
SAHI Lexical Only	55.0%	60.0%	11.0%	0.532	0.532	0.548
SAHI Dense Only	60.0%	65.0%	12.0%	0.525	0.538	0.556
SAHI Hybrid + RRF†	**65.0%**	**65.0%**	**13.0%**	**0.477**	**0.519**	**0.519**